Kidney Cyst Gene Expression Classifier

Classical (SVM-RBF) and quantum (QSVM) classifiers trained on GSE7869 (21 samples, 5 classes). Both scored 90.48% under nested Leave-One-Out CV — see the paper for the full evaluation. This app runs the FINAL models, trained on all 21 samples, for actual use, not a repeat of that evaluation.

Input must be RMA-normalized expression values already on the same scale as the training data (log2, quantile-normalized), with probe IDs as column names. This app does not perform RMA itself — see the module docstring in app.py for why single-array RMA isn't meaningful without a frozen reference.

Model

- [ ] SVM-RBF (classical, recommended)
- [x] QSVM (quantum kernel)

Upload a CSV: one row per sample, columns = probe IDs

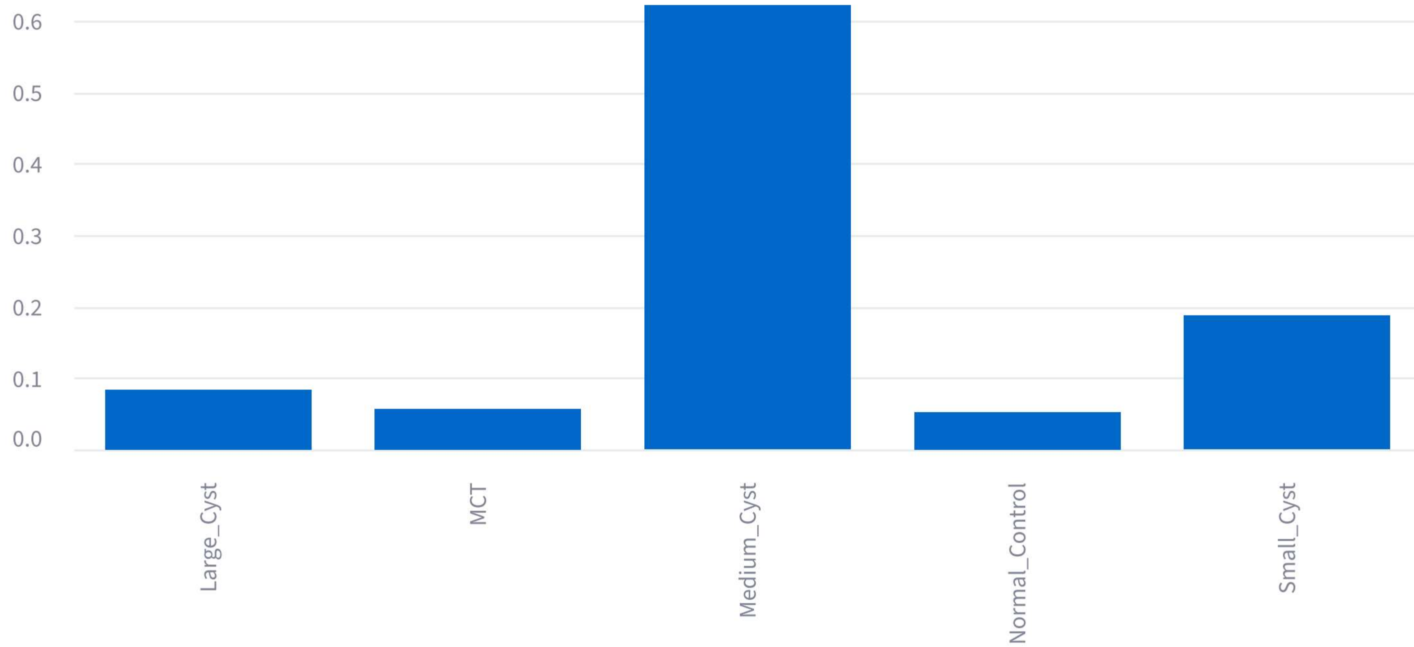
test_upload.csv
1.4MB

+

Loaded 1 sample(s), all 75 required genes present.

Classify

Sample 1: Medium_Cyst



Class	Probability
Medium_Cyst	0.621
Small_Cyst	0.187
Large_Cyst	0.0834
MCT	0.0566
Normal_Control	0.052

Limitations: n=21 training samples, several likely from the same 5 kidneys (patient-level leakage risk, not corrected here). Predictions on samples very different from the training distribution (different platform, tissue prep, batch) should not be trusted.